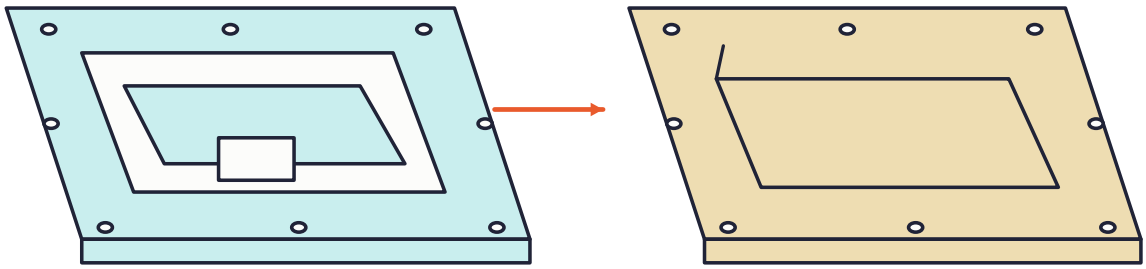
Make the soft parts

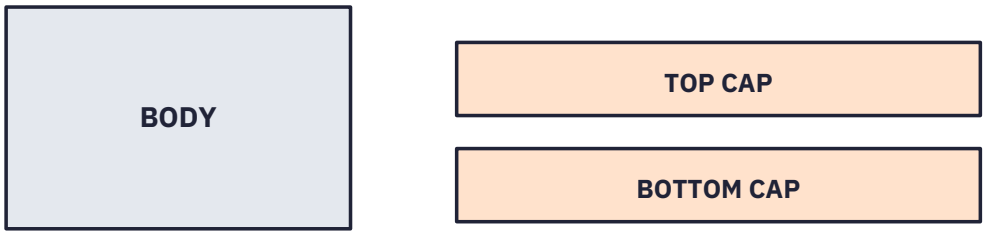
Silicone funnel + two foam caps + one body pour. A picture for each bench move.

FUNNEL



cavity + core + steel rod

THREE COLD-CORE POURS



1 Funnel: pages 2-16

Finish the printed tooling, prove the material stack, cast, open and inspect the removable funnel.

2 Cold core: pages 17-22

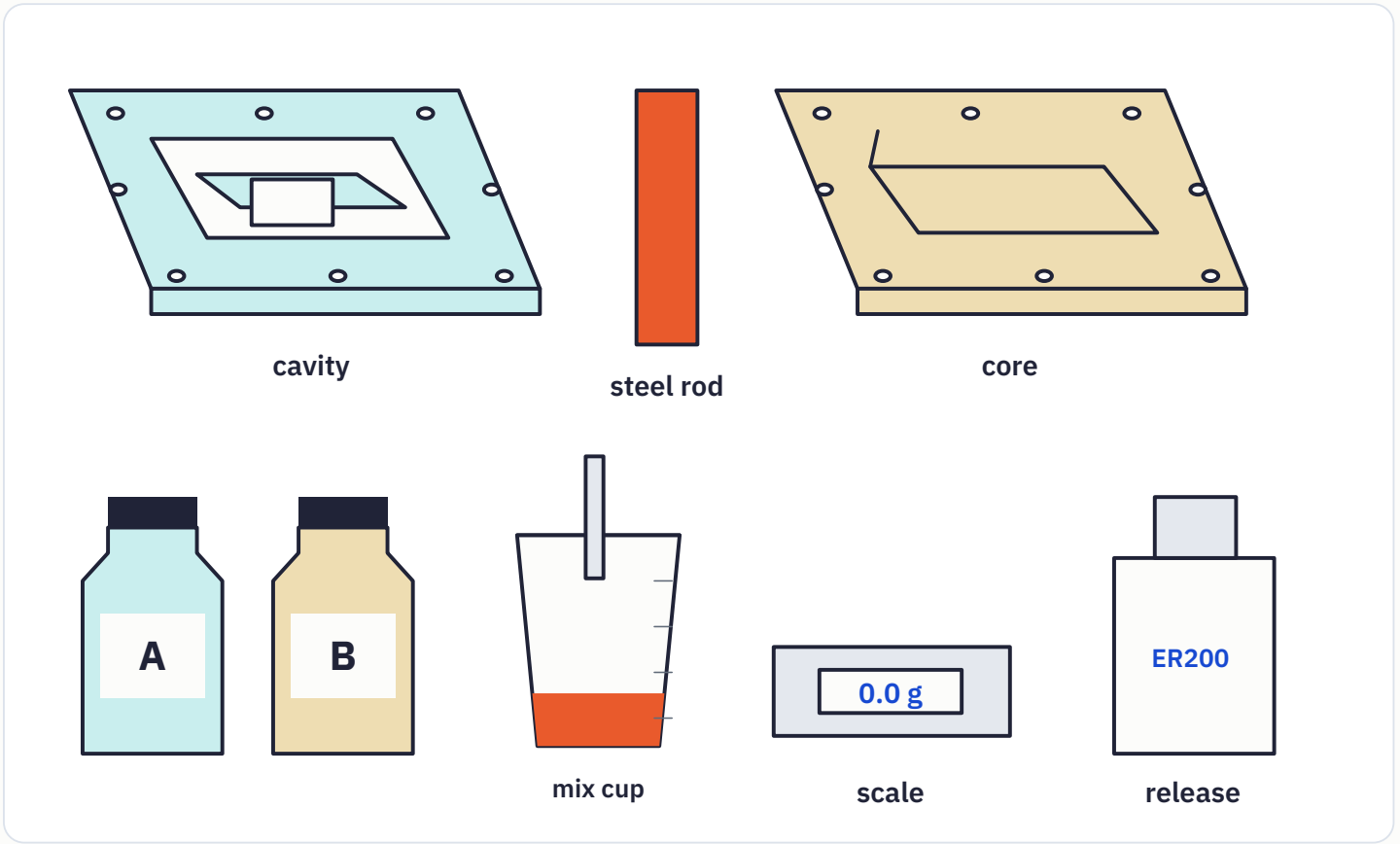
Two cap pours bolted to the shell's top face, then the body pour through its open top. Each mixes 1:1.

USE THIS AT THE BENCH

Coral marks the current move. Blue marks gauges, air paths and surfaces to keep clear. Pictures are schematic; named dimensions govern.

Stage the casting bench

Lay out the whole job before silicone A meets silicone B.



1 Tooling + closure

Cavity, core, stock steel rod, eight M4 x 20 bolts, 9 mm OD washers and nuts; a sharp blade for cured flash.

2 Liquids + handling

BBDINO 40A A/B, compatible black pigment, scale, cups, sticks, release, catch tray, vacuum chamber and pump.

3 Dry, ventilated work

Read the container instructions. Wear eye protection and appropriate gloves; keep release spray away from ignition sources.

READY

Room, materials and tooling are at the recorded batch temperature. The open mold and catch tray fit through the actual chamber opening.

Tooling: funnel-mold/README.md. Release: Smooth-On ER200 instructions.

Finish the forming faces

The coating allowance belongs to the forming surfaces. The parting lands remain bare.



also mask: locators, rod guides and axial stops

1 Clear supports and open infill

Remove bolt-pocket supports. Drill 1.5 mm dry-side breathers: cavity sides 3.2 mm deep, core flat back 1.8 mm deep. Positions: mold README.

2 Measure a witness

Use the actual PETG, sanding and finish stack on a sample. Measure net growth; the shell reserve is 0.30 mm normal to the face.

3 Mask closure datums

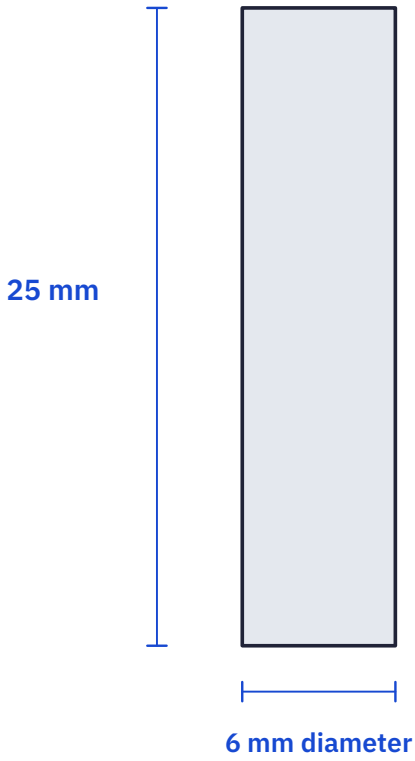
Keep lands, pegs/holes, bolt bores, rod guides and end stops bare. Coated lands change closure height.

BEFORE CONTINUING

The finish is dry and coherent, and the coated halves still close on the bare lands. Coating compatibility and closure need a physical check.

Prepare the steel rod

One stock 6 mm x 25 mm rod forms the entire straight outlet bore.



304 STAINLESS STEEL

smooth stock cylinder

clean + light release film

ends stay outside the cast

1 Select the stock rod

Use the specified smooth 6 mm x 25 mm stainless dowel. Its factory end chamfers stay outside the silicone forming span.

2 Clean the cylinder

Remove oil and debris, then dry. Keep the cylindrical surface smooth and free of dents or raised burrs.

3 Apply release

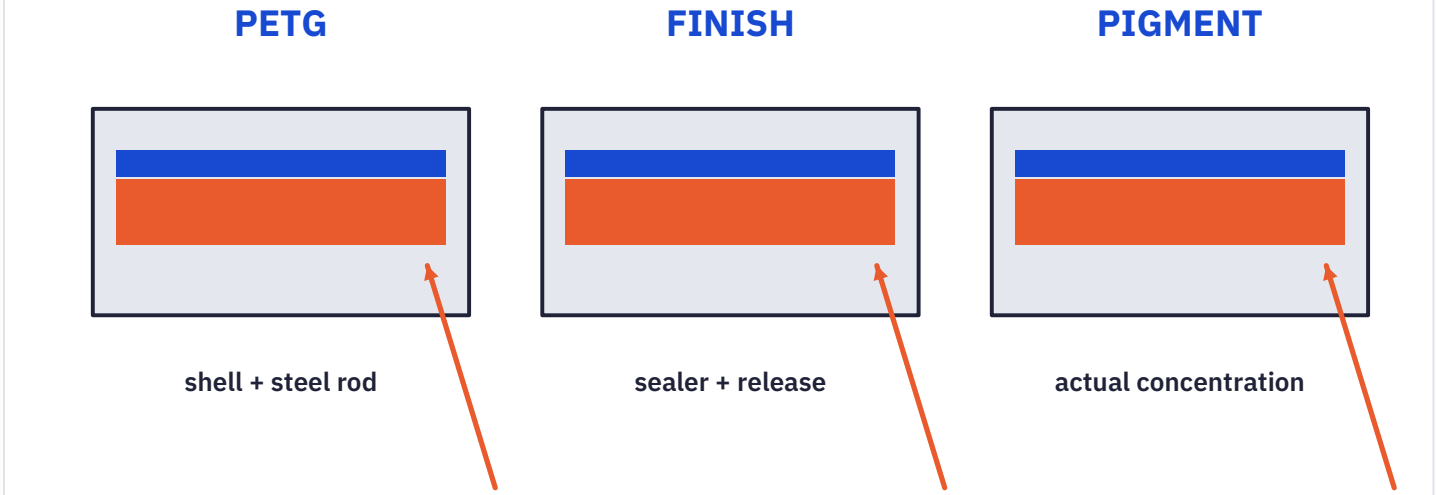
Apply the same light release film used on the mold. Keep the rod and both printed guides clean for a loose drop-in fit.

READY

The straight rod slides through the open core guide and rests on the cavity's lower floor. Its upper end stays visible.

Prove the material stack

A small same-stack witness protects the first full casting.



actual 40A silicone + actual pigment on each actual contact stack

cure, then peel and inspect the contact face

1 Make the witness representative

Use the actual shell PETG and finish, steel rod, release and silicone batch. Include the planned pigment concentration.

2 Keep the stack clean

Use clean cups, tools and contact surfaces. Test the exact finishing products and pigment that will touch this platinum-cure silicone.

3 Inspect after cure

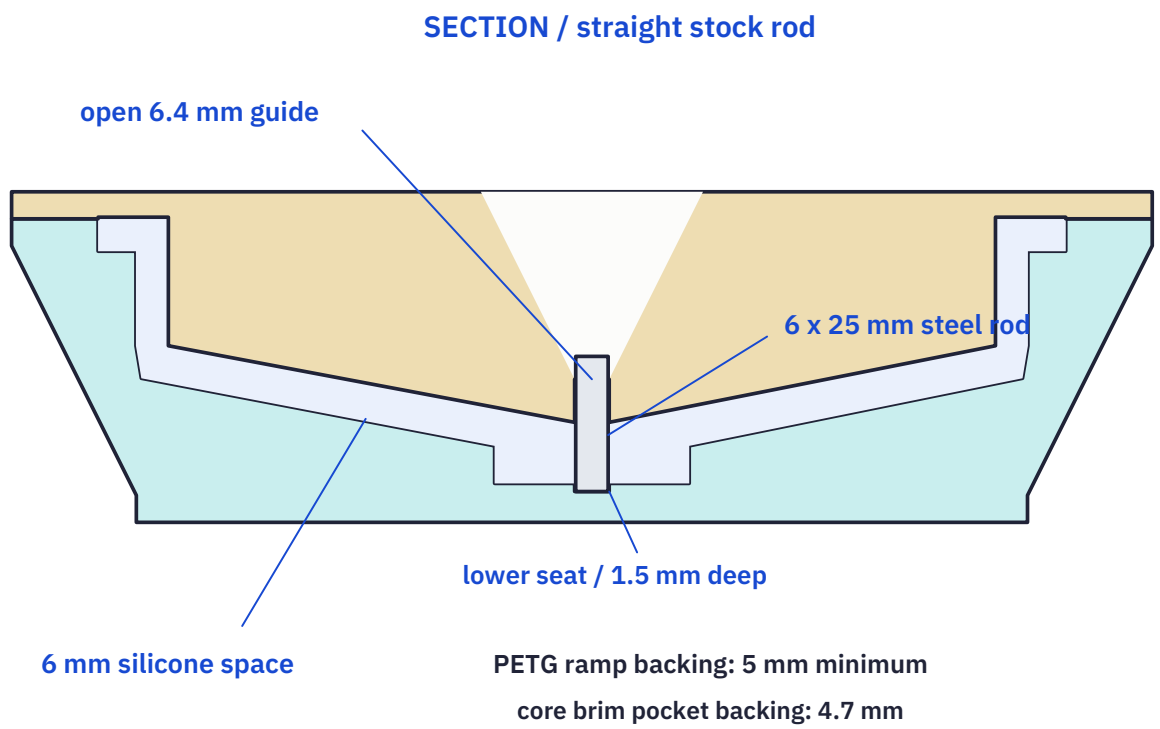
The rubber must cure at the contact face and release without tack, tearing or coating transfer. Record that result before the full cast.

HOLD IF GUMMY

A firm outer surface can hide inhibited rubber at the tool. Cure/release compatibility has no accepted full-tool record yet.

Drop in the steel rod

The open core guide leads the rod into the cavity's lower floor seat.



1 Close the empty mold

Seat the locators and bare flange lands. The core guide must align with the cavity's lower seat.

2 Drop the rod through the guide

Lower the stock rod from the open dry back. It rests on the lower floor, 1.5 mm below the block-bottom face.

3 Check free movement

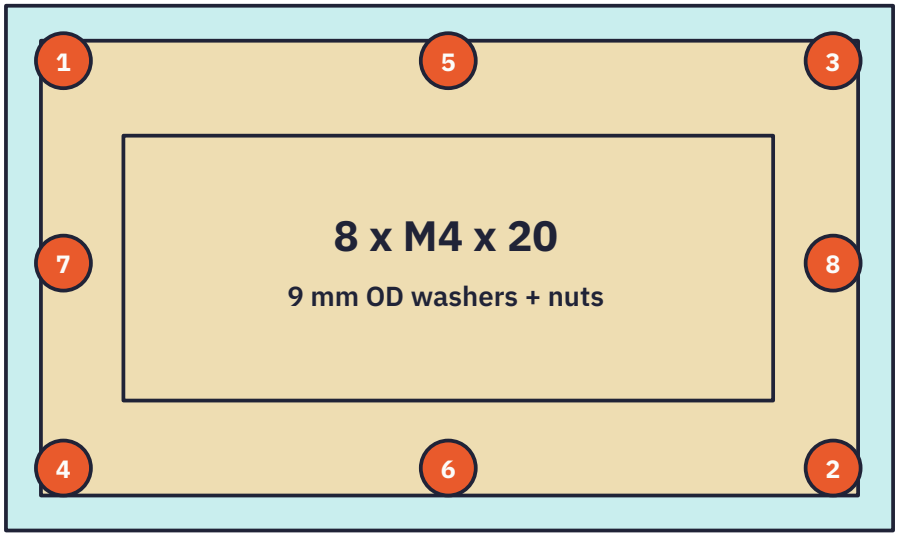
Lift the rod and let it settle again without pressing. The upper end projects about 4.2 mm above the guide.

READY

The mold closes on its bare lands independently of rod length. Clear support residue if the rod catches.

Dry-close the mold

Bring the bare parting lands together before adding liquid.



opposite stations / small increments

bolt order schematic; actual asymmetric locators set orientation

1 Locate without force

The asymmetric short pegs establish drain orientation. One mating hole is slotted to accommodate spacing error.

2 Close opposite stations

Use eight bolts, washers and nuts or small clamps on the flat flange backs. Tighten incrementally until bare lands meet.

3 Backlight the seam

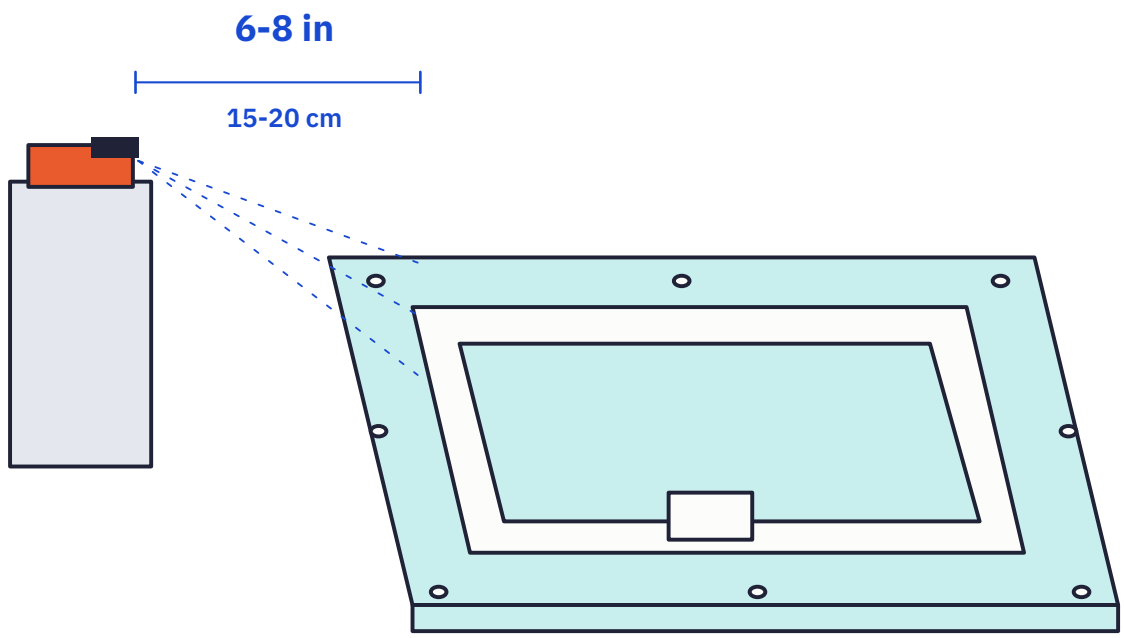
Check closure with the steel rod resting on the lower floor. Reopen for release and filling after this dry fit succeeds.

CHECK

The bare flange lands meet, the rod rests on its lower floor, and the open upper guide leaves its end accessible.

Apply a light release film

A thin, dry, continuous film reaches the small wet features too.



mist / brush detail / mist / dry

1 Clean and shake

Use a suitable cleaner for the proven finish stack. Shake Ease Release 200 well; spray with ventilation and eye/skin protection.

2 Mist from 6-8 inches

Apply lightly, brush over fine details and add a second light mist. Coat the steel rod lightly; keep the guide and fill/vent holes open.

3 Let it dry

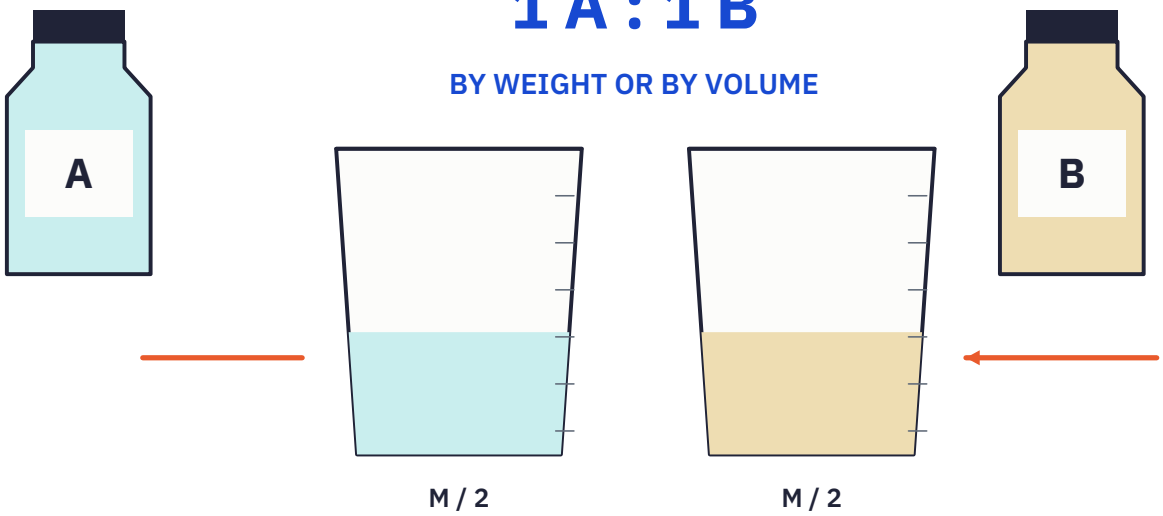
Recheck bare closure datums, rod guides and port openings. Apply a light fresh release film before each cast.

RELEASE CHECK

The same-stack witness must already have passed. Spray application does not establish compatibility, cleanability or finished food-contact acceptance.

Measure, color and mix

BBDINO 40A uses equal A and B. Choose one measurement basis for the whole batch.



331.0 mL nominal casting

about 374 g rubber + ports and mixing allowance

1 Set a batch quantity

Nominal geometry is 331.0 mL. The ~1.13 g/mL material estimate gives ~374 g in the casting; allow extra for fill ports and mixing loss.

2 Weigh the pigment

Use the actual BBDINO platinum-compatible black. Project limit: at most 2% by weight. Include it in the same-stack cure/release witness.

3 Mix the measured components

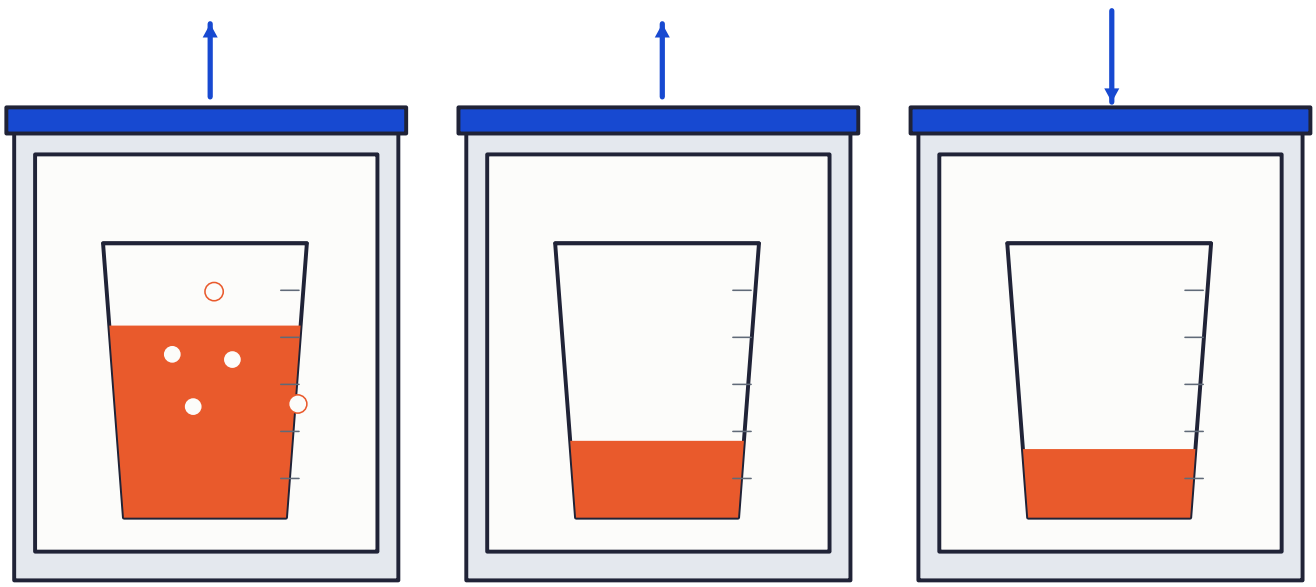
Scrape the cup wall and bottom until uniform. Start the working-time clock when A and B meet; no unmixed streaks remain.

BATCH CLOCK

Maker working time: 30 min at 23 C. Record room and liquid temperature; warmer conditions shorten the window. Follow the supplied batch instructions.

Degas the mixed silicone

This 40A grade requires degassing. Give the rising foam room above the liquid.



headroom above actual batch / catch tray below

1 Choose an ample cup

Mixed silicone rises during evacuation. Use a larger container or split the batch if the available cup cannot contain the froth.

2 Evacuate under control

Degas in a separate container with expansion space. Observe the rise and collapse; control the valve to prevent overflow.

3 Vent and pour while fluid

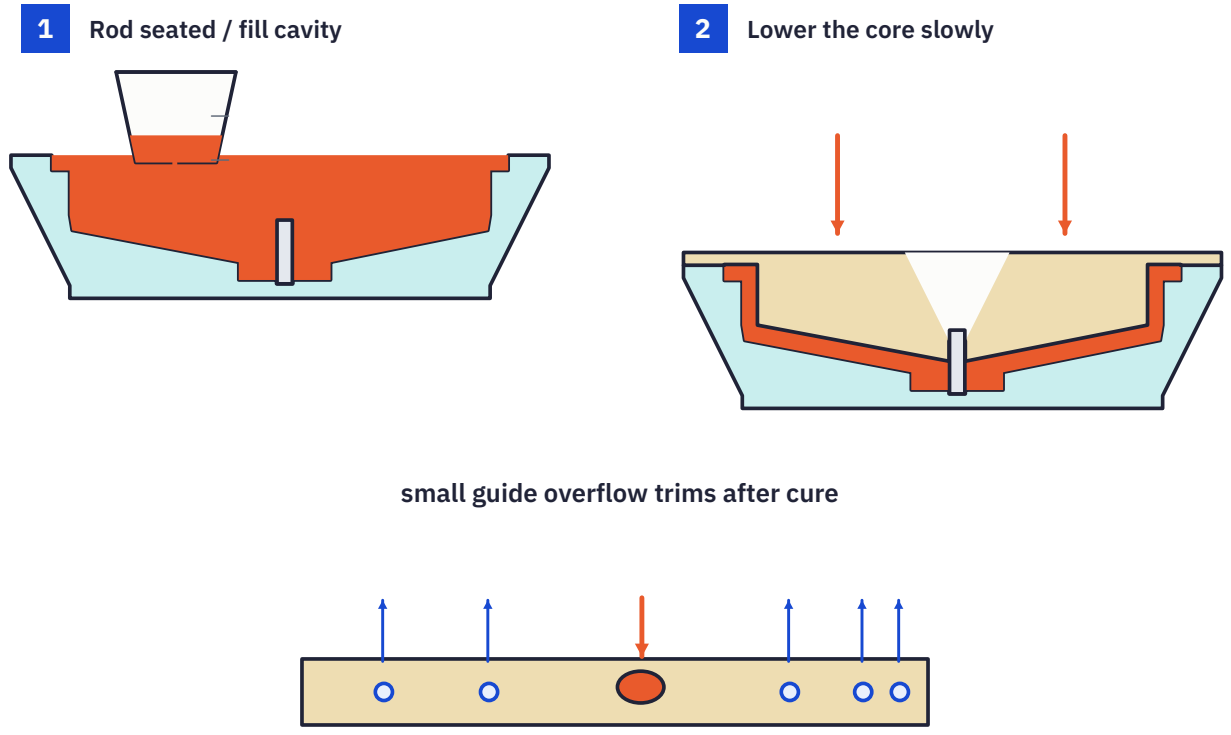
Vent slowly, remove the cup and continue the casting sequence within the measured working-time window.

USE THE ACTUAL EQUIPMENT INSTRUCTIONS

The guide supplies no universal vacuum setpoint or timer. Follow the chamber/pump and silicone batch instructions; watch the material.

Fill, lower, then top up

Set the rod in the lower seat, fill the open cavity, then lower the core slowly over it.



1 Seat the rod, then fill

Set the steel rod on the lower seat floor. Fill around the rod and drain block while the degassed silicone still flows.

2 Seat the core evenly

Lower slowly over the rod. Let air escape through the open rod guide; close the bare flange lands without force.

3 Top up through the fill hole

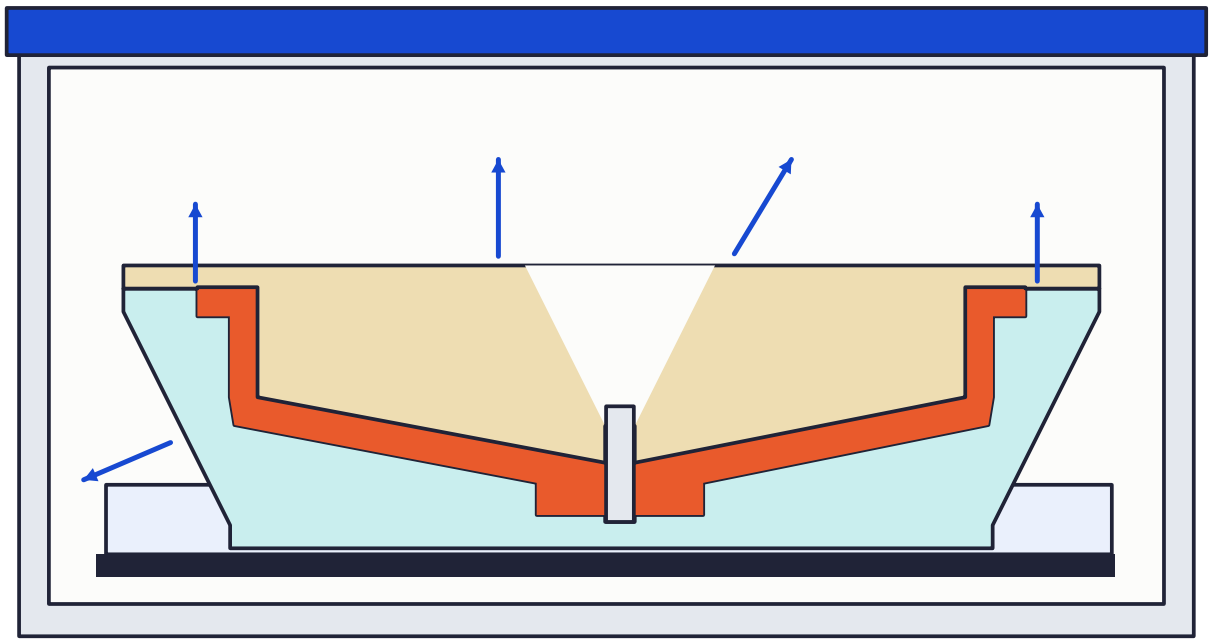
Use the 11 mm fill opening. The five 4 mm vents, rod access and dry-side infill breathers remain open; catch overflow.

CHECK

The parting lands remain closed, the rod stays seated, and liquid reaches the block and continuous 6 mm ramp space. No pressure injection.

Cycle the filled mold gently

If using a filled-mold vacuum cycle, the whole tool shares the chamber pressure.



fill + vents + rod access + infill breathers open

catch tray / overflow clear of pressure paths

1 Keep openings connected

Check the assembled mold and catch tray through the chamber's actual opening. Keep fill, vents, rod access and drilled breathers open.

2 Evacuate and vent slowly

Perform any cycle while silicone is fluid. Observe tool/overflow behavior, vent gently, then recheck and top up the fill level.

3 Return to ambient for cure

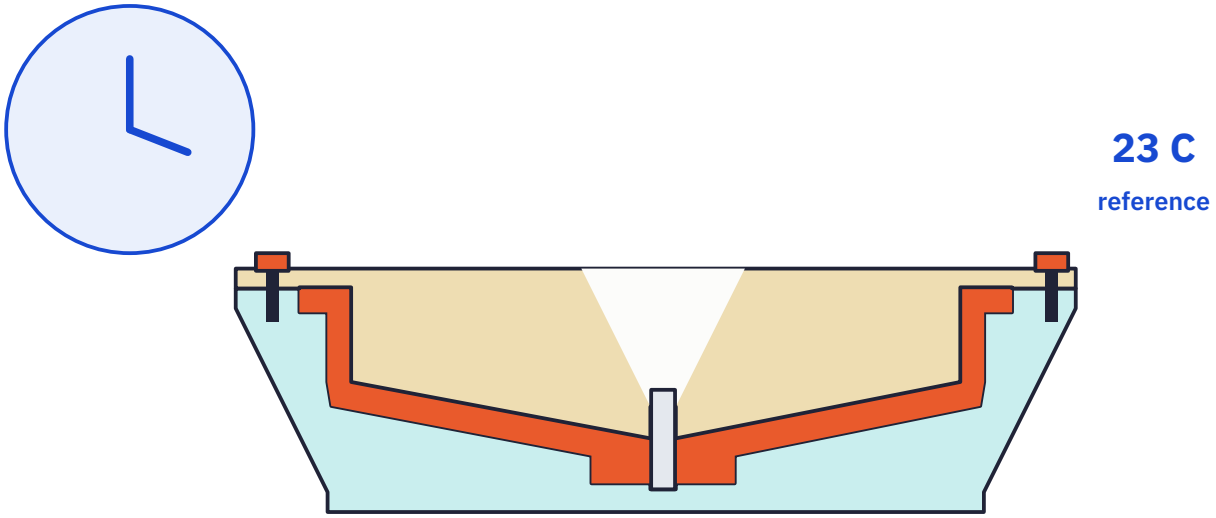
Keep the flanges held together through room-temperature cure. The tooling is not rated for a sealed atmosphere differential.

KEEP PRESSURE PATHS OPEN

Liquid head is about 0.6 kPa. Six walls and 15% gyroid rely on open infill breathers during slow vacuum cycling. Stop on distortion or a blocked path.

Hold the flanges through cure

Leave the tool undisturbed at ambient pressure while the silicone develops its cure.



23 C
reference

5 h project demold hold

24 h project full-use hold

also inspect a same-batch witness

1 Log time and temperature

Record A/B mixing start, pour finish, room/liquid temperature and batch identity. Warmer and cooler shops change the cure window.

2 Use the conservative project hold

Project material record uses 5 h before demold at 23 C and 24 h before full use. Check the supplied batch insert and cured witness.

3 Keep the tool assembled

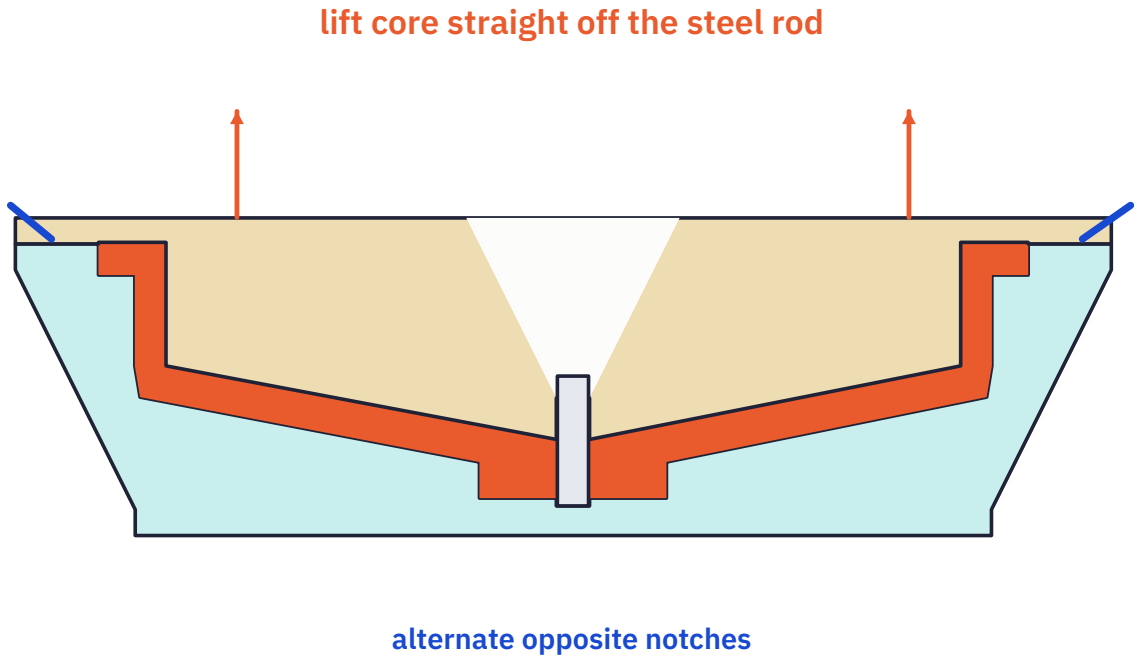
Maintain ambient pressure and closed flanges. Leave the casting and steel rod seated until their contact surfaces have cured.

CURE IS NOT FOOD-CONTACT RELEASE

The manufacturer's direct page lists 3 h; the project holds 5 h conservatively. No qualified post-cure bake schedule is established.

Open in small movements

Free the core first; keep the casting and steel rod together while peeling from the cavity.



1 Clear the opening points

Trim overflow at port mouths and remove the flange bolts or clamps. Keep the casting supported while freeing the core.

2 Alternate opening points

Use a blunt tool at opposing edge notches, moving a little at a time. Peel accessible silicone brim to let air enter.

3 Lift, then peel

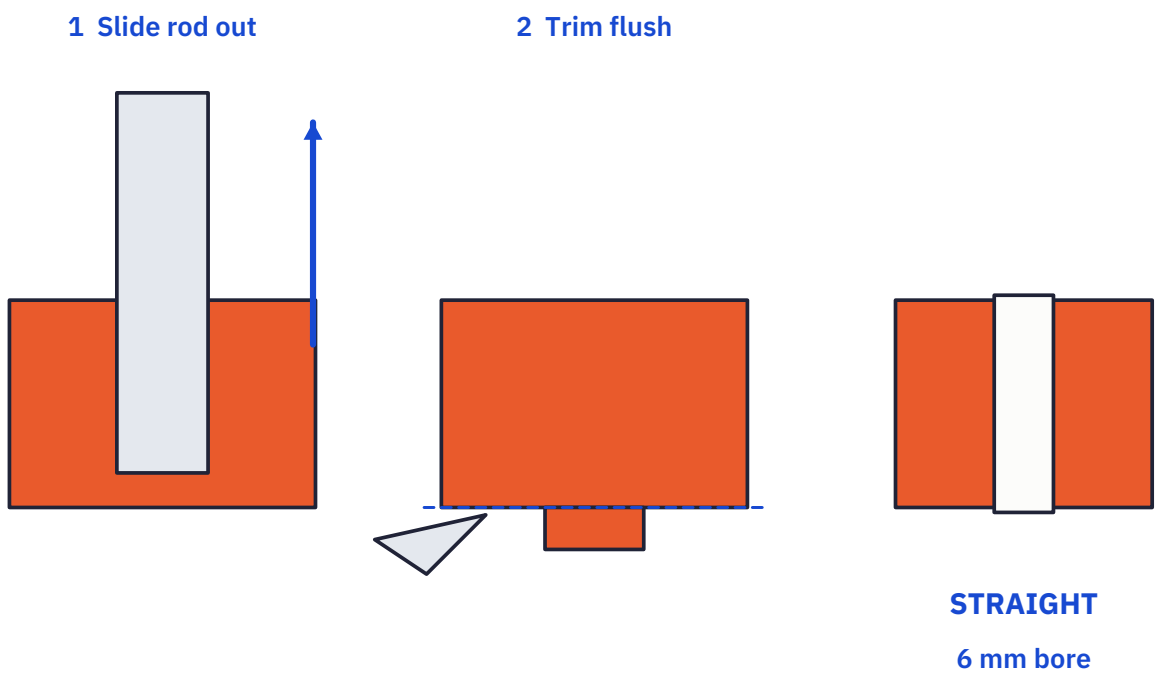
Lift the core straight off the rod. Peel the casting and rod together from the cavity; support the drain block.

STOP ON STICKING

Peel the accessible brim to admit air; keep opening movements small. Support the drain block without levering on the rod.

Pull the rod and trim flash

Support the silicone block while sliding the straight steel rod out.



1 Withdraw the straight rod

Support the block and slide the released steel rod out along its axis. Peel the silicone from it if needed.

2 Trim the accessible collars

Cut lower-seat flash flush with the block bottom and guide overflow flush with the bowl throat. Trim port and vent flash.

3 Keep the bore intact

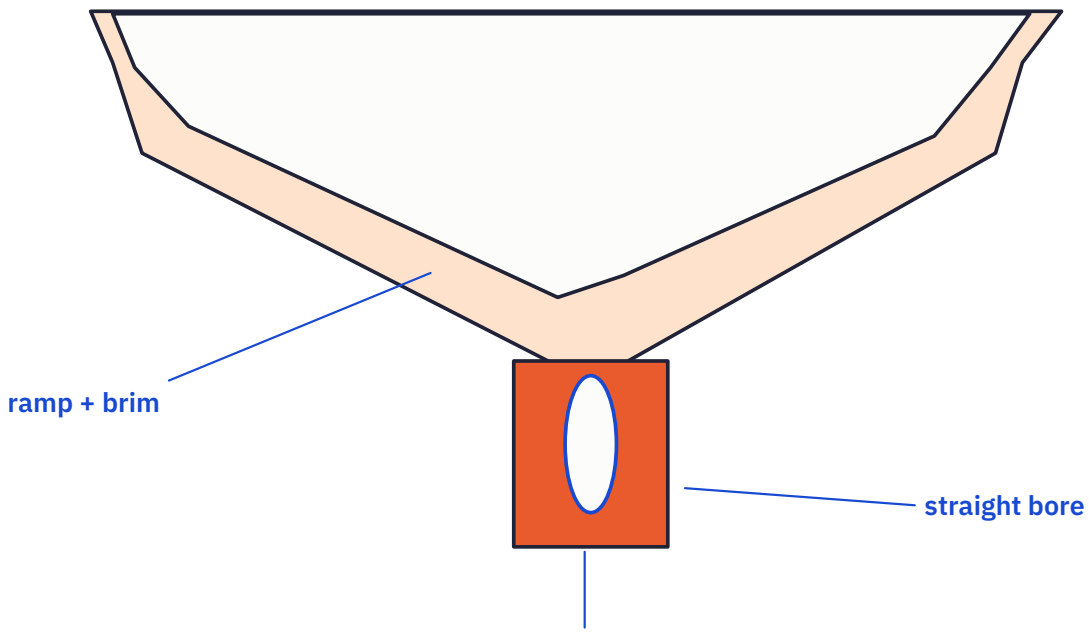
Leave the cylindrical 6 mm wall smooth. Clean the rod and guide openings before the next casting.

READY

The straight outlet is open end to end, its edge is clean, and the flat bearing face is intact.

Inspect and clean the funnel

A complete casting has a continuous ramp, an open straight bore and an intact bearing face.



1 Inspect every wetted surface

Look for pinholes, bubbles, tack, tears and coating transfer. Inspect the straight bore and both trimmed outlet ends.

2 Clean with the qualified process

Release traces and residue must be removed. The finished silicone, black pigment and post-process need the wetted-surface qualification.

3 Check its actual installation

The block bears on the cradle's hook tops. Its 6 mm bore grips the 6.35 mm LLDPE drain stub. Check seating, lift-out and leaks.

HOLD FROM FOOD SERVICE

No qualified bake schedule or finished-mixture food-contact result is recorded. A cured, attractive part alone does not close that gate.

Prepare the three foam pours

Two cap cavities and one body cavity use the acquired FSD 2 lb closed-cell PU foam.



MIX 1:1

measured shot per pour
top cap / 12.6 mm deep
bottom cap / 16 mm deep
body / open top, all at once
~5 L risen foam per build

TOP / BOTTOM / BODY

1 Use the acquired foam

FSD B08R7TX8QJ two-part 2 lb closed-cell PU. Mix the two parts 1:1 in measured shots, one shot per pour.

2 Stage three separate pours

Each cap pours bolted to the shell's top face; the body pours last through its open top. A build fills ~5 L of risen foam (~4 L cavity + waste), about 1/7 of the 1 qt kit.

3 Stage a dry, ventilated bench

Use dry cups/tools, protective gloves and chemical eye protection. Keep moisture out of the liquids and vapors away from the operator.

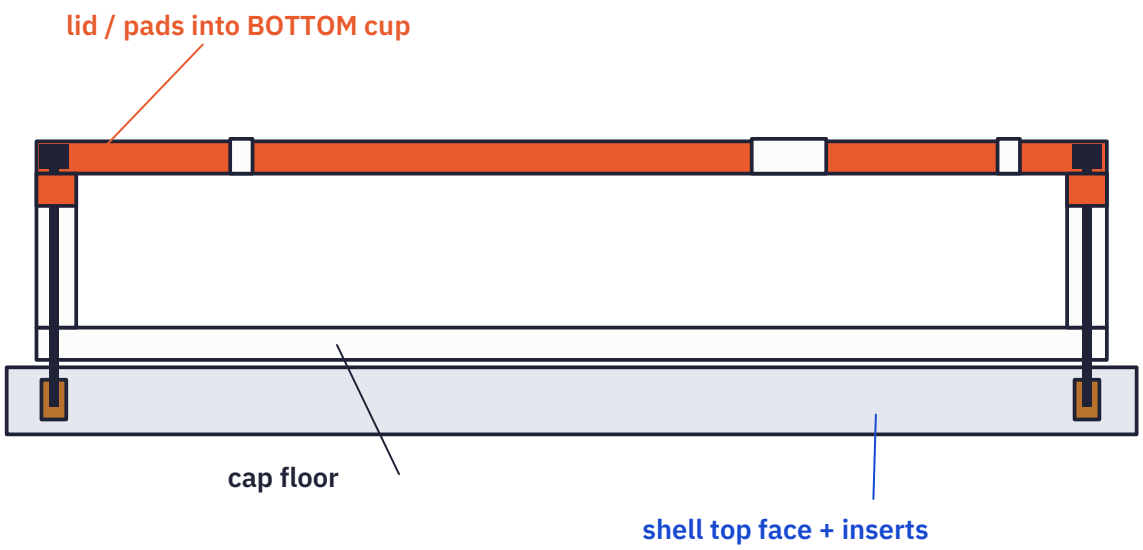
GLOVES ON

The PU foam's isocyanate component is a skin sensitizer. Clamp each cap before mixing its shot.

Cold-core.md steps 3 and 6; BOM foam row; FSD official foam page + SDS

Clamp each cap mouth-up

Use the shell's top face as the pour fixture for both cap stacks, one after the other.



TOP: flat lid underside / 4 deck inserts first

BOTTOM: counterbore pads face into the cup

1 Label TOP and BOTTOM

The top cap carries conduits and four deck-mount columns. Press the four full-length RX-M3x5.7 inserts into those columns before pouring.

2 Use installed shell threads

Ten full-length M3 inserts in the shell's top face supply the pour-clamp threads. Lid and cap have clearance holes.

3 Build the correct stack

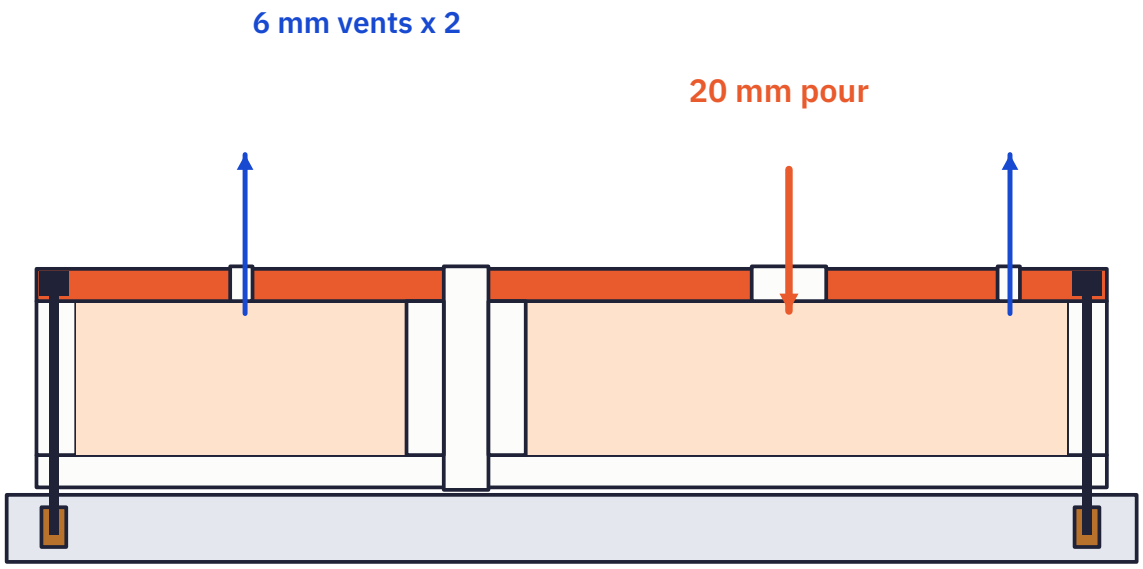
Cap floor down on the shell. Bottom lid pads enter the cup; top lid underside is flat. Close with ten M3 x 25 screws.

CHECK

The lid is held by the complete screw pattern, not its own threads. All top-cap conduits and both vent paths remain open.

Pour the two cap cavities

Liquid enters through the lid's 20 mm hole and air leaves through both vents.



TOP 12.6 mm / BOTTOM 16 mm pour depth

top-cap section / one stack at a time

1 Mix 1:1

Mix the two parts 1:1 with the clamped cap ready beside you, and pour as soon as the shot is mixed.

2 Pour into the 20 mm hole

Keep the two 6 mm vents clear while foam expands. The lid stays clamped; do not cover the vent paths to force fill.

3 Cure, trim, unbolt, repeat

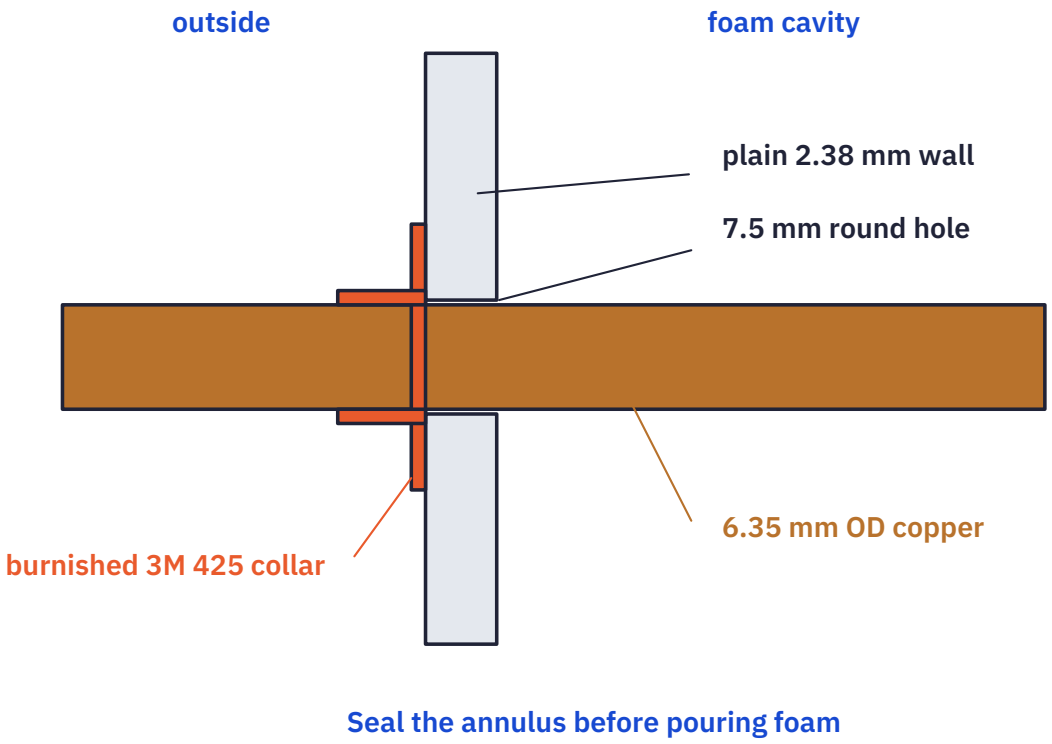
After cure, trim pour/vent overflow to the plate, never into a cradle. Remove the ten screws and repeat for the other labeled cap.

PRESERVE THE TOP FEATURES

Trim flush without cutting a valve cradle, conduit, anchor or deck-mount feature. Every conduit bore stays open end to end.

Seal the copper exits; protect access

Everything foam will bury is installed, routed and proved before the body pour.



Seal the annulus before pouring foam

1 Prove the buried work

Carbonator and inlet jet qualified; coil/probes/reeds and fitted fluid lines checked. Tug-test inaccessible push fittings before encapsulating.

2 Form, inspect and seal the exits

Feed bare tails through their holes while lowering the core. Support broad bends in the copper; reject kinks or flattening. Burnish foil collars to clean copper and shell around both annuli.

3 Keep service paths open

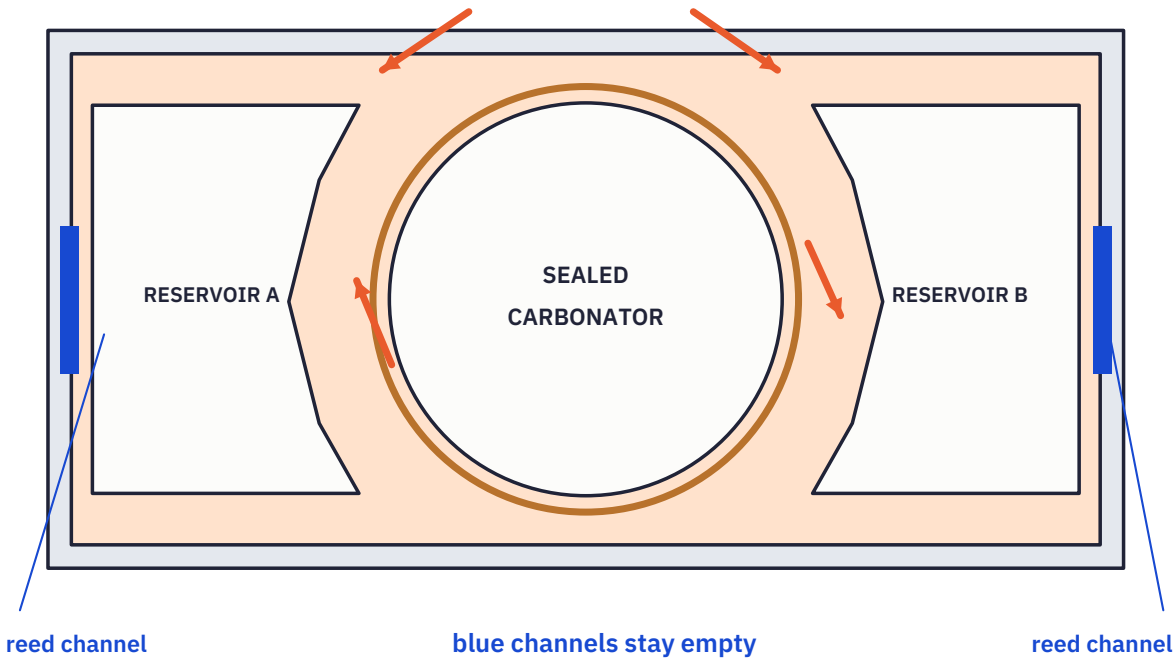
Protect reservoir vent/fill openings, top conduits, relief outlet and both reed channels. Reed columns are installed after body foam cure.

PROBE AND ACCESS CHECK

Both probes and lead entries need foam coverage without a trapped void. Do not bury an unproved sensor, line, relief path or fitting.

Pour through the open top

The body foam fills around the sealed carbonator and reservoirs; its connected zone reaches the coil.



CORAL = foam zone / sealed vessel interiors take no foam

1 Mix the body shot 1:1

Mix 1:1 and pour all at once into the open +Z top, with no lid and no down-channels. The cap stacks are already poured separately.

2 Pour directly into the body

One pour fills the connected outer gap and space around the carbonator/coil, through the open paths at the reservoir-pocket ends.

3 Watch fill and squeeze-out

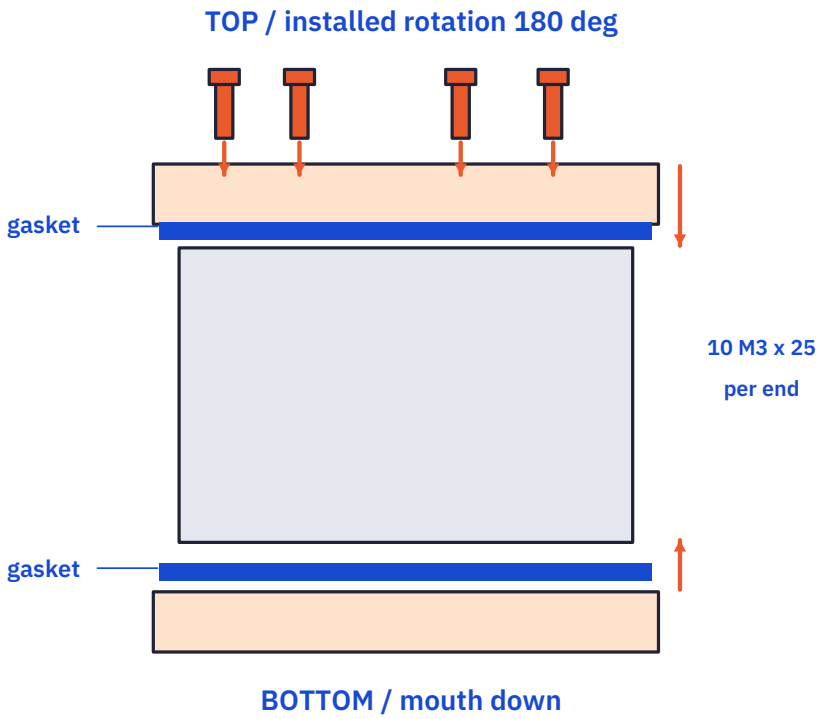
Preserve empty reservoir/reed cavities and open tube paths. Beads squeeze out at the lane slots and tight tube exits; trim them after cure.

A VOID AT A PROBE IS THE DEFECT

The pour must fully wet each probe and its lead entry. Trapped air condenses and frosts on the cold metal.

Trim and close the cold core

All three pours are cured before the gaskets and caps complete the insulated stack.



1 Trim only cured overflow

Trim to the plate, never into a cradle. Protect tubes/leads, copper exits, cradles and insert hosts; clear all conduits, vents and reed channels.

2 Install reeds, gaskets and caps

Drop reservoir reed columns in, route their cables out the top, and lay both TPU gaskets. Top cap rotates 180 deg; bottom mouth faces down.

3 Close and inspect

Ten M3 x 25 screws per end. Identify top orientation by deck/cradle features. Check open paths, gasket seating and recessed screw heads.

FINISHED

Both caps and body cured; exposed foam flush; probes wetted in with no void; channels and relief path open.